



Overview of Electrical Engineering

Electrical Engineers design systems that have two main objectives:

1. To gather, store, process and present information.
2. To distribute, store and convert energy between various forms.

Major Classifications:

a) **Communication systems** generate, transmit and distribute information.

Examples: telephone systems, radar systems, satellite systems,...

b) **Computer systems** process and store information in digital form. Examples: A typical modern automobile contains tens of special purpose computers. Many electronic devices, household appliances, chemical processes and railroad switching yards routinely controlled through computers.

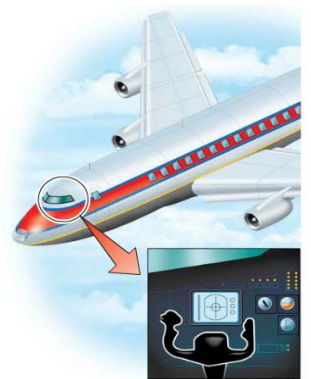
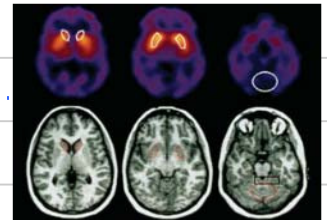
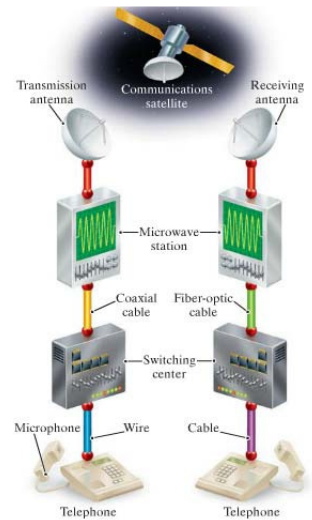
c) **Signal processing systems** transform signals and the information contained in them into a the useful form.

Examples: Image processing, voice recognition, fault detection,...

d) **Power systems** generate, transmit and distribute electric power. Electric power is generate in large quantity in power plants by nuclear, hydroelectric and thermal (coal-, oil-, or gas fired) generators. Power is transmitted and distributed across the country by transmission lines and power grid.

e) **Control systems** gather information with sensors and use electrical energy to control a physical process.

Examples: Temperature control in heating/cooling system, temperature, pressure and flow rate control in oil refinery, autopilot systems in airplanes,...



All these branches have something in common which is **Electric Circuit Theory.**

Fundamental Concepts

Electrical phenomena are defined based on the concept of electric charge.

Characteristics of electric charge:

1. Charge is bipolar.
2. Charge exists in discrete quantities, which are the integral multiples of electronic charge. (1.6×10^{-19} coulombs)
3. Electrical effects are attributed to both separation of charge and charge in motion.

Voltage is the energy per unit charge created by separation of positive and negative charges.

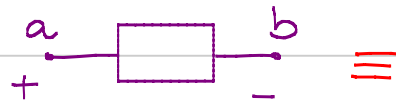
$$V = \frac{dw}{dq}$$

w : energy in joules

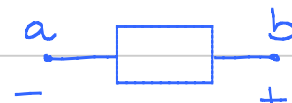
q : charge in coulombs

V : voltage in volts.

$$1 \text{ V} = 1 \frac{\text{J}}{\text{C}}$$



$$V_{ab} = 5\text{V}$$



$$V_{ba} = -5\text{V}$$

Current is the rate of charge flow.

$$i = \frac{dq}{dt}$$

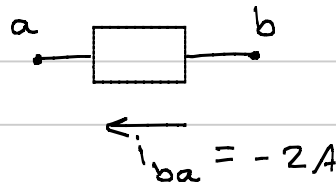
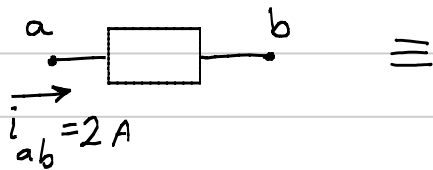
i : current in amperes

q : charge in coulombs

t : time in sec.

$$1 \text{ A} = 1 \frac{\text{C}}{\text{s}}$$

There are two types of current: 1. direct current (dc)
2. alternating current (ac)



Ex. The current flowing through an element is

$$i = \begin{cases} 2 \text{ A}, & 0 < t < 1 \\ 2t^2 \text{ A}, & t \geq 1 \end{cases}$$

Calculate the charge entering the element from $t=0$ to $t=2$ s.

$$i = \frac{dq}{dt} \rightarrow q = \int i dt$$

$$q = \int_0^2 i dt = \int_0^1 2 dt + \int_1^2 2t^2 dt$$

$$= 2t \Big|_0^1 + 2 \cdot \frac{1}{3} t^3 \Big|_1^2 = 2(1-0) + \frac{2}{3}(8-1) = \underline{6.67 \text{ C}}$$

Ex. The total charge entering a terminal is $q = 10 - 10e^{-2t}$ (mc). Find the current at $t = 0.5$ s.

$$i = \frac{dq}{dt} = \frac{d}{dt} (10 - 10e^{-2t})$$

$$= 0 - (-20e^{-2t}) \Big|_{t=0.5} = \underline{7.36 \text{ mA}}$$

Ideal circuit element:

1. It has only two terminals .
2. It is described mathematically in terms of its voltage and current.
3. It cannot be subdivided into other circuit elements.

Passive sign convention (or, Passive Reference Configuration): when the current enters through the positive terminal of an element, passive sign convention is satisfied. In this case, use a positive sign in any expression that relates the voltage to the current. Otherwise, use a negative sign.



PSC is satisfied

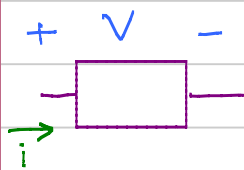
Power and Energy

$$P = \frac{dw}{dt}$$

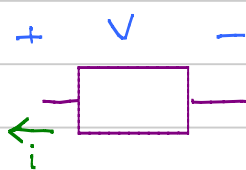
P : power in watts
 w : energy in joules

$$P = \underbrace{\frac{dw}{dq}}_V \cdot \underbrace{\frac{dq}{dt}}_i \Rightarrow \boxed{P = V \cdot i}$$

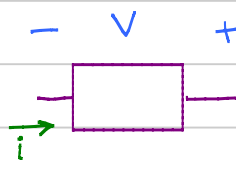
Applying P.S.C



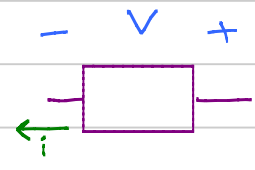
$$P = Vi$$



$$P = -Vi$$



$$P = -Vi$$

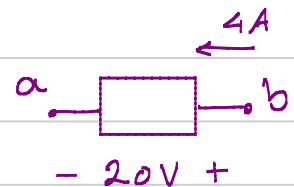


$$P = Vi$$

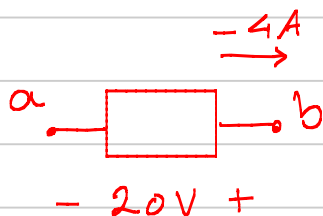
If $P > 0$, the circuit element absorbs power and if $P < 0$, the circuit element delivers power.

In any circuit sum of the element powers must equal zero.

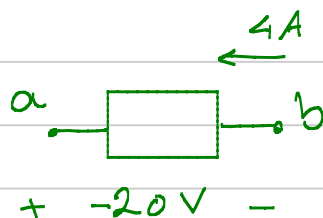
Ex. Assuming a 20 V drop from terminal b to a, and a current of 4 A enters terminal b.



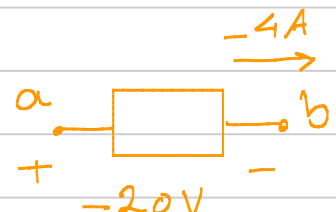
$$P = Vi = 20 \times 4 = 80 \text{ W (abs.)}$$



$$P = -Vi = -(20)(-4) = 80 \text{ W (abs.)}$$



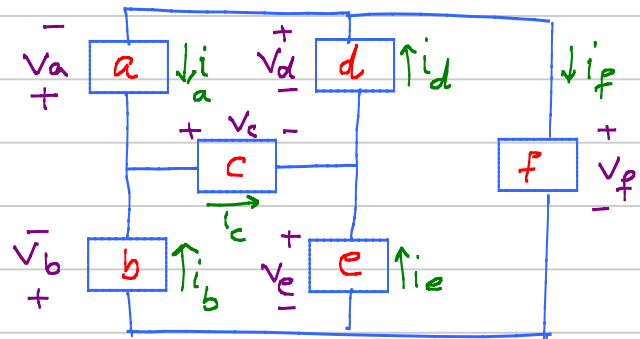
$$P = -Vi = -(-20)(4) = 80 \text{ W (abs.)}$$



$$P = Vi = (-20)(-4) = 80 \text{ W (abs.)}$$

Ex. The numerical values of voltages and currents are given in the table. Find the power of each element.

element	Voltage	Current
a	-18	-51
b	-18	45
c	2	-6
d	20	-20
e	16	-14
f	36	31



$$P_a = -Vi = -(-18)(-51) = -918 \text{ W (del.)}$$

$$P_b = Vi = (-18)(45) = -810 \text{ W (del.)}$$

$$P_c = Vi = (2)(-6) = -12 \text{ W (del.)}$$

$$P_d = 400 \text{ W (abs.)}, P_e = 224 \text{ W (abs.)}, P_f = 1116 \text{ W (abs.)}$$

Power balance check: $\sum P \stackrel{?}{=} 0$

$$-918 - 810 - 12 + 400 + 224 + 1116 = 0 \quad \checkmark$$